SUPPLEMENTAL READING FOR PROGRAMMING ASSIGNMENT 1 PART 3

IMAGE MORPHOLOGY: TOP HAT

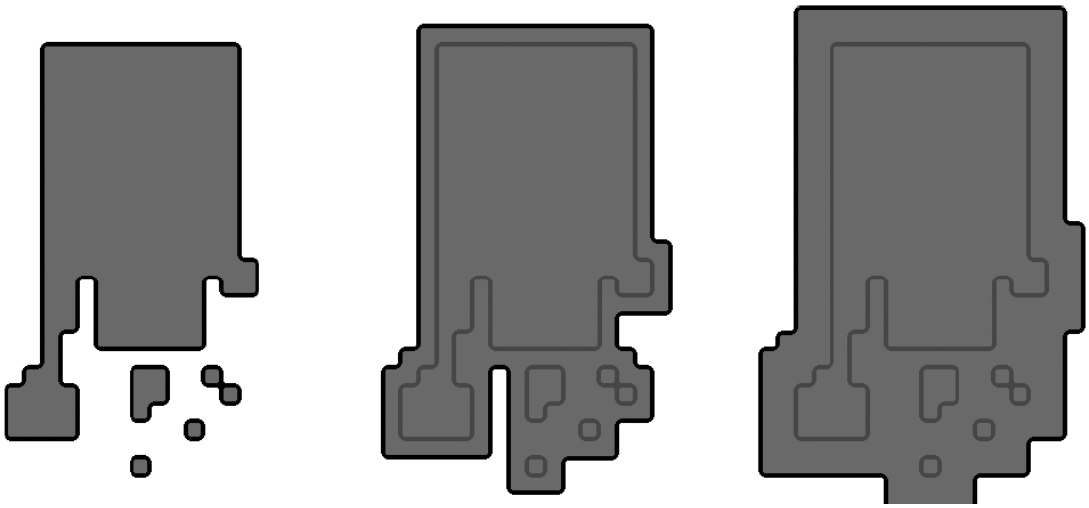
Morphological transformations are very popular in image processing and in computer vision where they are mostly used as tools for preprocessing images: removing small elements, closing holes in images, detecting edges among many other operations.

The programming assignment is about using Top Hat, which isolates brighter areas of an image. This operation is based on Morphological Opening, which in turn is itself a combination of Erosion and Dilation operators.

This material will provide you with a quick and visual understanding of all underlying concepts. These are all implemented and available from OpenCV python library cv2.

1. Morphological Dilation

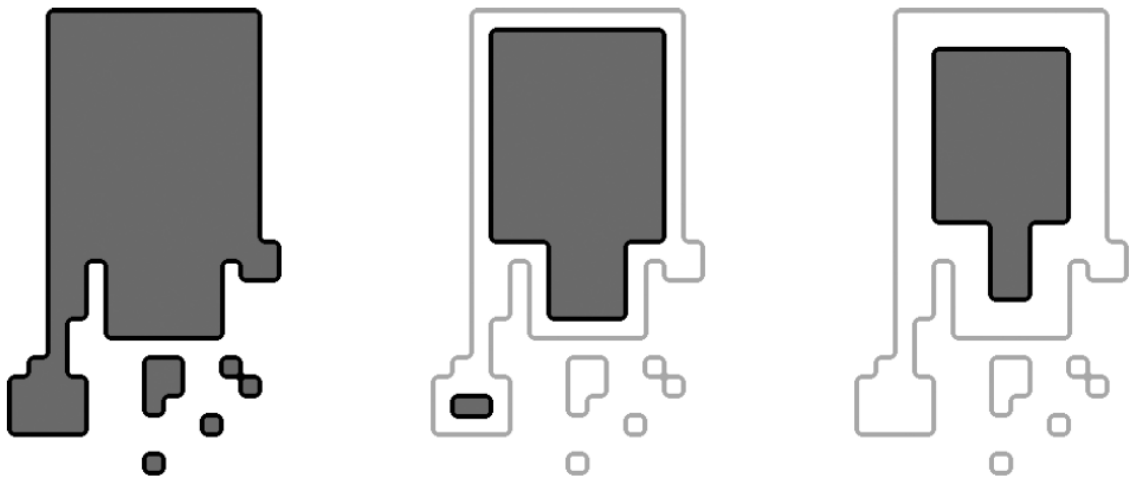
This operation will take the maximum under a square kernel.



(a) Original Image (b) Dilation (c) Dilation Twice

2. Morphological Erosion

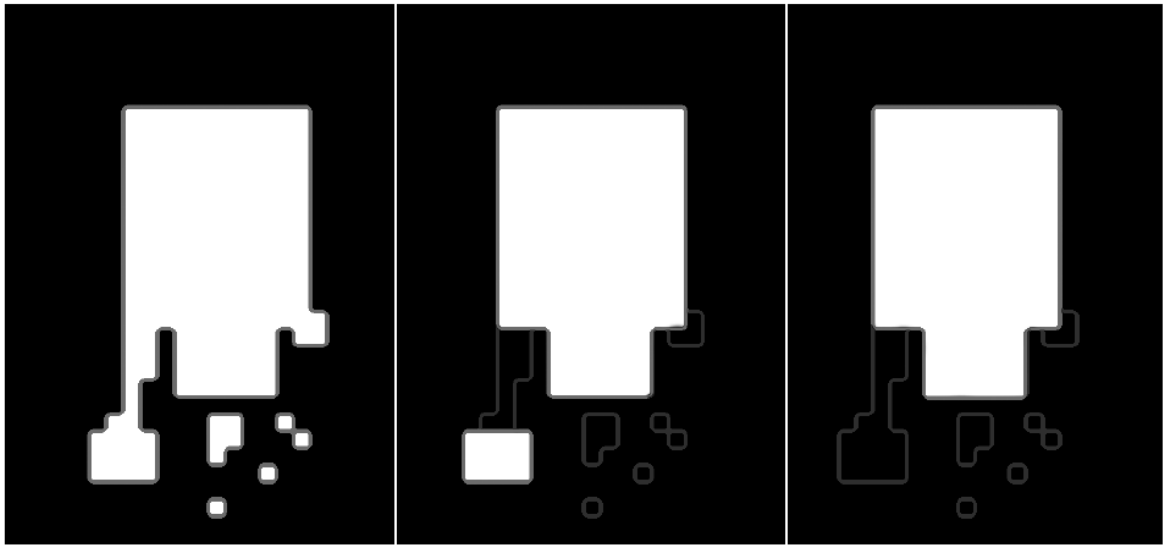
This operation will take the minimum under a square kernel.



(a) Original Image (b) Erosion (c) Erosion Twice

3. Morphological Opening

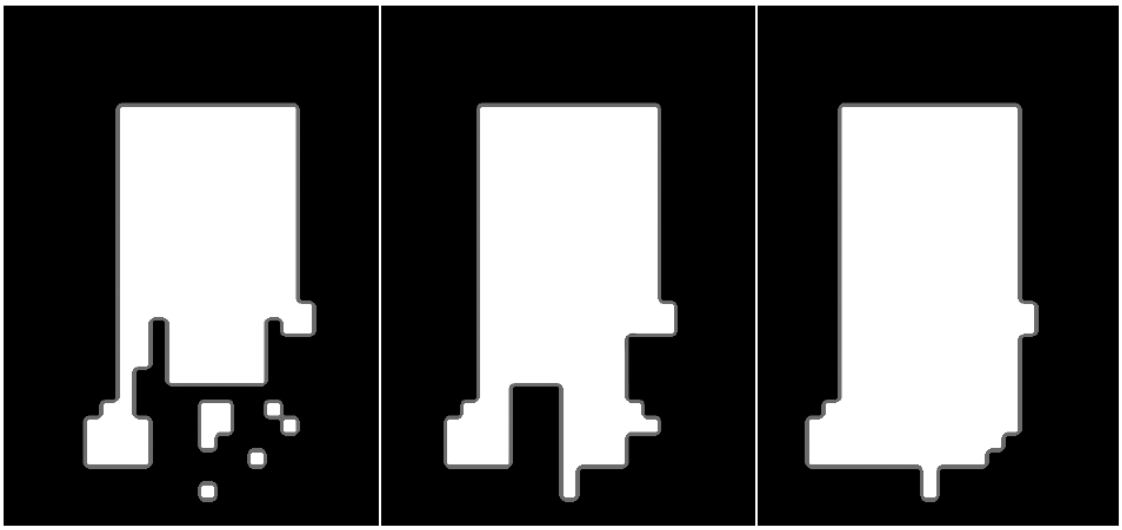
The Opening operator is illustrated in pictures below.



(a) Original Image (b) Opening (c) Opening Twice

4. **Morphological Closing**

The Closing operator is illustrated in pictures below.



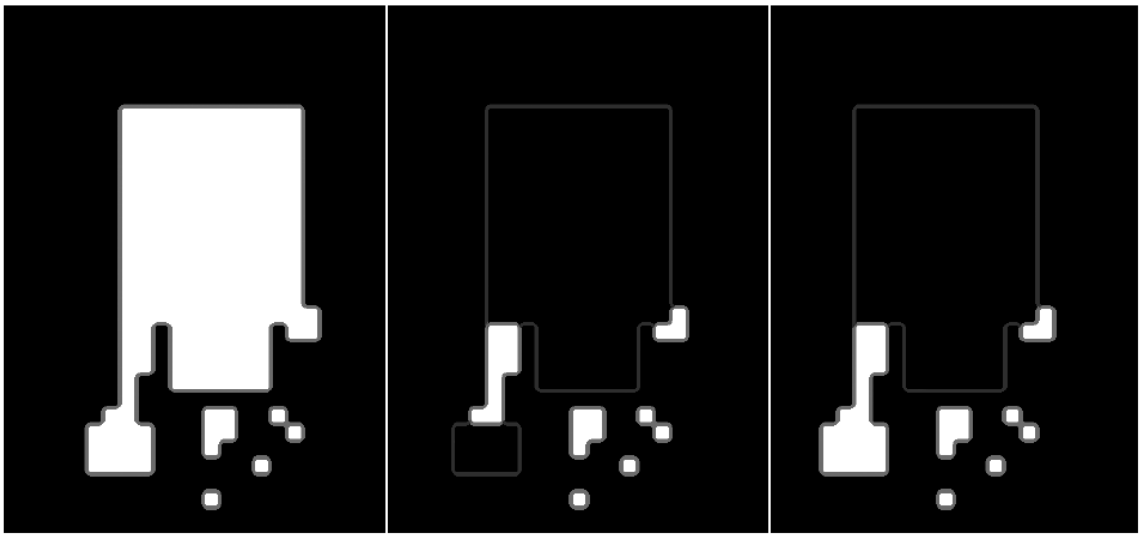
(a) Original Image (b) Closing (c) Closing Twice

5. **Top Hat**

The Top Hat operator is used to isolate patches of the image that are brighter than their immediate neighbors. It will be more useful when trying to isolate parts of an image with brightness changes relatives to the object they belong to.

TopHat(source_image) = source_image – Open(source_image)

The Top Hat is a subtraction of the Opened version of the source image from the source image itself.
As described earlier, the Opening operator highlights small cracks or local drops in the original image. Therefore, by subtracting Open(source_image) from the source_image, reveals areas that are lighter tan the surrounding region of the source image, relative to the size of the kernel element used in the Opening operator.



(a) Original Image (b) Top Hat (c) Top Hat Twice

6. **Application to Programming Assignment 1 Part 3**

   a) Here you are only required to run the Top Hat operator on the original image and display the result as an image

   b) The result of the Top Hat operator is still a grayscale image showing brighter spots in the range 0-255. To create a mask, you will have to change it into a binary image where the brighter spots are thresholded to 255 value and the darker spots are clipped to 0 value.

   That binary (0, 255) image is a mask.

   This image is called a mask because it is used to only let through corresponding pixels in the original image where the mask has non-zero values.

   c) In this part of the exercise, you will use the mask created in section b) to apply it on the original image, but only let through the parts of the image where the corresponding pixel on the mask has the bright value of 255.

   The result will be an image showing only the brighter spots of your original image.

   Experiment with a few images and select the most interesting results that you would be proud to have on your report.